Electrostatic–hydrophobic cooperation in the mesoscale assembly of synapsin and synaptic vesicles

Short title: Force control of synapsin–vesicle assemblies

Shokoofeh Akbari[1], Oleg Shupliakov[2], Yousef Jamali[1,*]

[1]Department of Applied Mathematics, School of Mathematical Sciences, Tarbiat Modares University, Tehran, Iran

[2]Department of Neuroscience, Karolinska Institutet, Stockholm, Sweden

[*]Correspondence: y.jamali@modares.ac.ir

Yousef Jamali ORCID: https://orcid.org/0000-0002-5709-3821

Shokoofeh Akbari: shokoofe.akbari1997@gmail.com

Oleg Shupliakov: oleg.shupliakov@ki.se

Abstract

Synapsin associates with synaptic vesicles, but local association does not always connect vesicle groups. We used a three-dimensional Brownian agent model to compare electrostatic and hydrophobic interactions in a four-condition design. Particle abundance, confinement and dispersion attraction were fixed. Electrostatics alone placed synapsin near almost every vesicle, although most vesicles remained in small groups. Hydrophobic attraction alone brought more vesicles into the largest assembly, with less synapsin association. Together, the interactions produced a larger connected assembly than the sum of their separate effects predicted. More synapsin particles were close to multiple vesicles, and short-lag synapsin mobility was lower. Contact composition and network connectivity distinguished local vesicle coverage from collective assembly.

Author summary

Synapsin helps organize the vesicles that store neurotransmitters. We asked whether the interactions that bring synapsin close to vesicles also connect vesicles into larger groups. In a computer model, electrical interactions placed synapsin near almost every vesicle, but the vesicles remained in separate groups. Short-range hydrophobic attraction brought more vesicles into the largest group. When both interactions were active, many more synapsin particles were near several vesicles and a large connected assembly formed. Synapsin also moved more slowly. Counting proteins near vesicles and measuring connections between vesicles thus describe different aspects of the same assembly.

1 Introduction

A presynaptic terminal holds synaptic vesicles (SVs) in groups while allowing them to move and participate in neurotransmission. The protein-rich SV membrane offers many sites for local interactions [1]. As vesicles and proteins move, these interactions can also link vesicles into larger assemblies.

Experiments on synapsin have linked molecular interactions to vesicle organization. Early bilayer experiments identified both electrostatic association and hydrophobic engagement of the

synapsin head domain [2]. Purified-vesicle experiments distinguished lipid binding from an ionic-strength- and phosphorylation-sensitive protein-associated contribution [3]. Computational cage and cytoskeletal cross-linking models then examined how phosphorylation-dependent binding could redistribute vesicles [4]. Later, perturbation of an amphipathic lipid-packing sensor motif linked membrane curvature sensing to synapsin binding and vesicle clustering [5]. Binding and clustering therefore involve more than one type of interaction.

Reconstitution experiments showed that synapsin assemblies capture vesicles and disperse following kinase treatment [6]. Perturbing the disordered region of synapsin also affects vesicle clusters in a living lamprey synapse [7]. Single-molecule tracking linked vesicle sequestration to restricted synapsin motion [8]. The mechanism also depends on the synaptic context. Song and Augustine distinguished synapsin-dependent condensation at inhibitory synapses from tetramer-mediated cross-linking at excitatory synapses [9]. Reserve-pool tracking also implicated Synapsin 2a tetramerization in vesicle organization [10]. These findings motivate separate measures of protein association, vesicle connectivity and particle motion.

Microscopy and a binding-particle model related assembly size and morphology to the synapsin-to-lipid ratio and vesicle adhesion [11]. X-ray scattering and cluster simulations resolved noncompact SV–synapsin assemblies with close intervesicle contacts [12]. More recent X-ray microrheology of synapsin assemblies with colloid-supported lipid bilayers found concentration-dependent motion and relaxation [13]. Together, these studies measure assembly structure and dynamics at different scales.

Simulation approaches span several resolutions. Atomistic synapsin simulations coupled to experiments revealed an ATP-dependent conformational route to vesicle binding and oligomerization [14]. Residue-level models resolve charge and hydropathy along protein chains [15], while coarse-grained simulations of repetitive peptides show that the balance of electrostatic repulsion and hydrophobic attraction depends on sequence and solution conditions [16]. At the particle level, the synapsin–vesicle model of Alfken et al. addresses binding redistribution and finite assembly size [11]. An agent model can represent thousands of moving proteins and vesicles while retaining their identities and pair interactions. We use this resolution to test how electrostatic and hydrophobic interactions change collective organization at fixed abundance and confinement. Random-graph and continuum-percolation studies show how connectivity can change nonlinearly with the number

and organization of edges [17,18]. This framework lets us ask how local synapsin association relates to the wider vesicle population.

We vary electrostatic and hydrophobic interactions independently in a three-dimensional Brownian agent model. Particle abundance, confinement and dispersion attraction remain fixed. Species-resolved contact networks measure protein association and vesicle connectivity. Saved trajectories also allow a mobility comparison with published tracking data. The comparison shows how the two interactions shape local association and collective vesicle organization.

2 Results

The four conditions differ in the electrostatic (E) and hydrophobic (H) interactions. Particle population, confinement and observation time are the same. Each condition has three independent simulation runs. We first compare network structure and then examine contacts by particle type and synapsin mobility. The latter analyses were developed after the runs.

2.1 Combined interactions produce a larger connected assembly

With neither E nor H active, the late mean all-agent largest-component fraction was 0.0238. Enabling H alone raised it to 0.0990 and E alone to 0.1053. Enabling both raised it to 0.5528 (table 1 and Fig 1). We compared this value with the sum of the separate E and H effects, using E0H0 as the common background. For each seed, we calculated the interaction contrast as

$$\Delta_{EH} = (Y_{11} - Y_{10}) - (Y_{01} - Y_{00}). \tag{1}$$

The mean contrast was 0.3723. Its three seed values were 0.3590, 0.3132 and 0.4445. The joint response exceeded the additive reference in every seed.

The time courses show how this difference developed (Fig 2A). At 0.14 ms, the mean largest-component fraction ranged from 0.043 to 0.066 across conditions. By 14 ms, it reached 0.021, 0.083, 0.057 and 0.409 for E0H0, E0H1, E1H0 and E1H1, respectively. At 28 ms the same values were 0.020, 0.090, 0.099 and 0.610. The early decline in E0H0 reflects relaxation of contacts present after initialization. The E1H1 assembly continued to grow through 28 ms. Continuing the same

Table 1: Late-window structural outcomes at the primary 2-nm association shell.

Condition	All-agent largest component	Non-isolated Syn	SVs in largest component
E0H0	0.0238 (0.0017)	0.3025 (0.0046)	0.0812 (0.0084)
E0H1	0.0990 (0.0071)	0.5157 (0.0034)	0.1839 (0.0153)
E1H0	0.1053 (0.0209)	0.6989 (0.0015)	0.1273 (0.0238)
E1H1	0.5528 (0.0802)	0.8831 (0.0028)	0.6111 (0.0886)

Entries are means across three independent seeds with between-seed SD in parentheses. E and H denote the electrostatic and hydrophobic kernels; van der Waals attraction is present in every row.

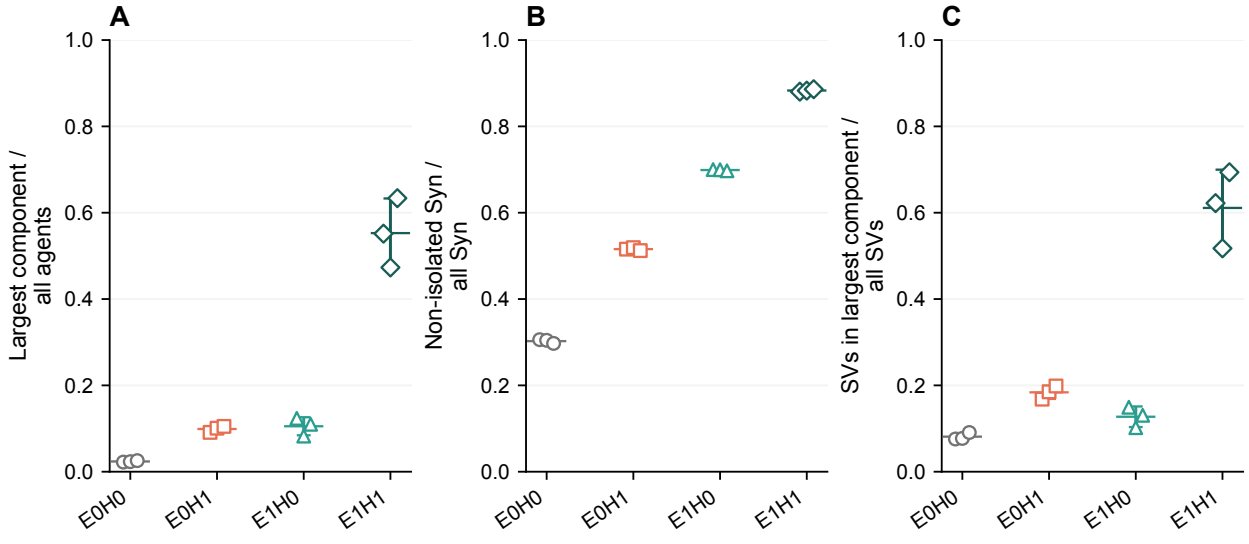


Fig 1: Late-window responses to the electrostatic (E) and hydrophobic (H) interactions at the 2-nm graph shell. Open symbols show individual runs; horizontal marks and bars show the mean and one between-seed SD ($n = 3$ per condition). Non-isolated synapsin includes Syn–Syn and Syn–SV components. Van der Waals attraction is active throughout.

three E1H1 trajectories to 84 ms showed markedly slower growth of the largest component at later times (Fig 2C).

2.2 Protein association and vesicle participation respond differently

Electrostatics produced the larger increase in the non-isolated synapsin fraction Q_{Syn}. With E alone, it reached 0.6989, compared with 0.5157 with H alone (table 1). The E-by-H interaction for this measure was −0.0289. Synapsin can belong to either Syn–Syn or Syn–SV components, so this measure covers both types of association.

The SV fraction in the largest component L_{SV} responded differently. It reached 0.1839 with H alone, 0.1273 with E alone and 0.6111 with both interactions (table 1 and Fig 2B). Its E-by-H contrast was +0.3811 and was positive in all three seeds (0.3805, 0.3075 and 0.4553). The conditional effects and their between-seed SDs are given in table 2.

The single-interaction comparison captures the main contrast: E1H0 had more participating synapsin than E0H1, but fewer SVs in the largest component. This ordering held in every seed. Local protein association and collective vesicle participation responded differently to the two forces.

Table 2: Paired seed-block contrasts at the 2-nm shell.

Outcome	Conditional contrast	Mean difference (SD)
All-agent largest component	H effect at E0	0.0752 (0.0056)
	H effect at E1	0.4475 (0.0687)
	E effect at H0	0.0815 (0.0212)
	E effect at H1	0.4538 (0.0783)
Non-isolated Syn	H effect at E0	0.2131 (0.0029)
	H effect at E1	0.1842 (0.0043)
	E effect at H0	0.3964 (0.0031)
	E effect at H1	0.3674 (0.0055)
SVs in largest component	H effect at E0	0.1027 (0.0089)
	H effect at E1	0.4838 (0.0744)
	E effect at H0	0.0461 (0.0249)
	E effect at H1	0.4272 (0.0846)

Values are mean differences in fractions, with the SD of the three paired seed differences in parentheses. "H effect" means H1–H0 at fixed E; "E effect" means E1–E0 at fixed H.

2.3 Local vesicle coverage and connections between groups

We separated Syn–Syn, Syn–SV and SV–SV contacts in the saved frames. At the 2-nm shell, the fraction of synapsin next to at least one SV rose from 0.1805 in E0H0 to 0.8775 in E1H1 (Fig 3).

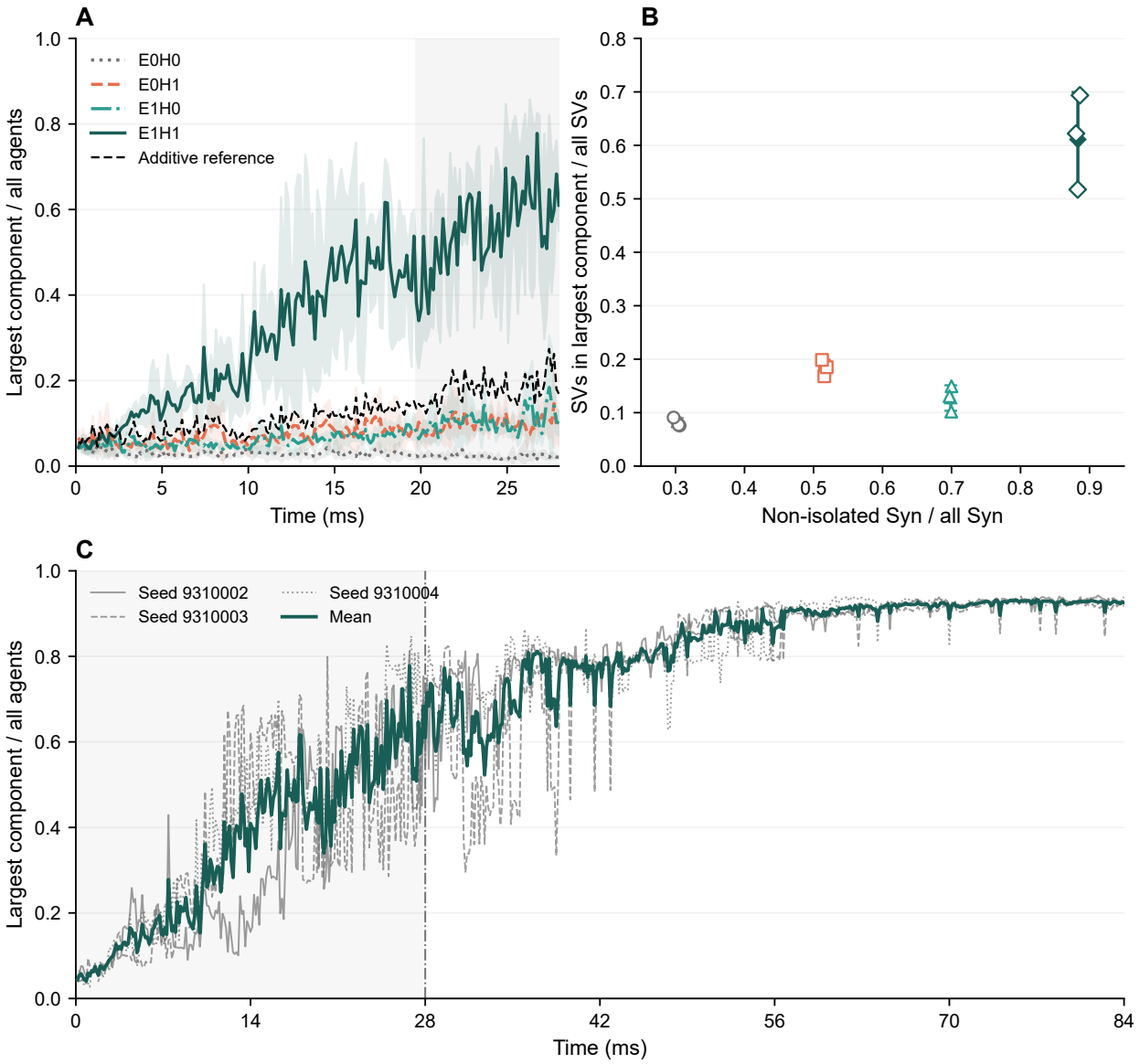


Fig 2: Network growth and vesicle participation at the 2-nm graph shell. (A) All-agent largest-component fraction. Lines show condition means, bands show one between-seed SD, and the dashed line is the additive reference $\overline{L}_{10}(t) + \overline{L}_{01}(t) - \overline{L}_{00}(t)$. Shading marks the final 60 frames. (B) Late-window non-isolated synapsin versus SV participation in the largest component. Open symbols show runs; filled symbols and bars show means and one SD. (C) Continuation of the three E1H1 runs to 84 ms. Gray lines show runs and the dark line their mean; the vertical line marks 28 ms.

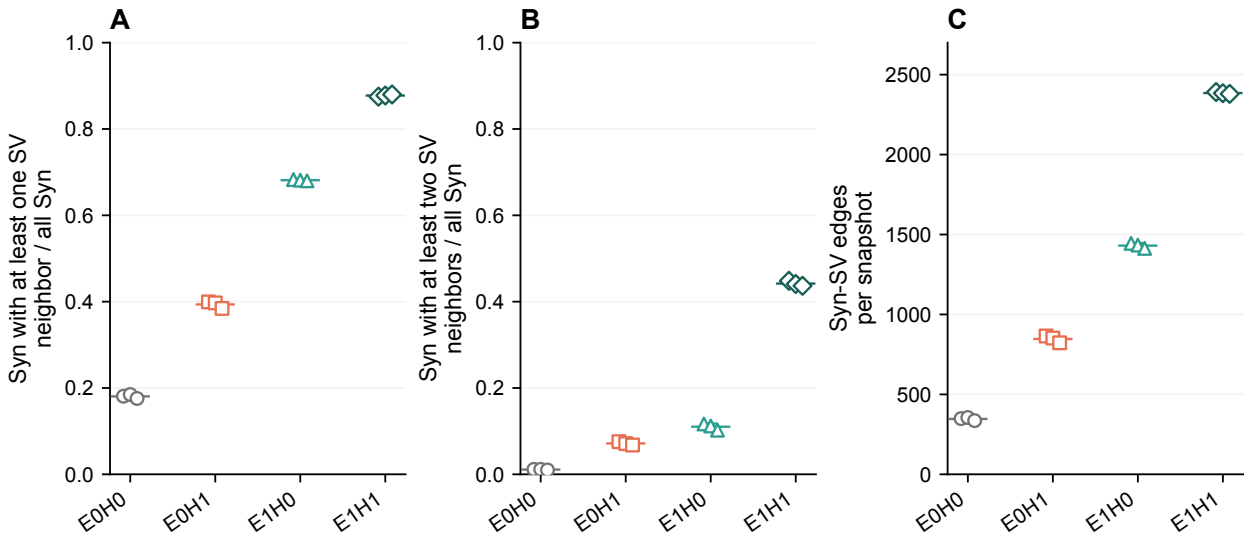


Fig 3: Species-resolved contacts at the 2-nm graph shell. (A) Fraction of Syn agents adjacent to at least one SV. (B) Fraction adjacent to at least two distinct SVs. (C) Number of Syn–SV edges per snapshot. Open symbols show individual late-window run means; marks and bars show the mean and one between-seed SD ($n = 3$ per condition). Contact uses outer radii, including the SV corona.

The corresponding E0H1 and E1H0 values were 0.3936 and 0.6814.

With E alone, 99.72% of SVs had at least one synapsin neighbor. This rose to 99.92% with E and H together. The fraction of SVs in the largest component nevertheless rose from 0.1273 to 0.6111. The fraction of synapsin next to at least two SVs rose from 0.1103 to 0.4419 between these conditions (Fig 3B). Mean Syn–SV edge counts per snapshot rose from 1,430.5 to 2,384.8, and Syn–Syn edges from 200.9 to 882.1.

The number of synapsin particles close to several vesicles helps explain how groups become connected, alongside the connected-component measures.

E1H0 also had more total edges than E0H1 (1,647.8 versus 1,277.2 per snapshot), although its largest component contained fewer SVs. It had fewer Syn–Syn edges (200.9 versus 299.1) and SV–SV edges (16.3 versus 131.5). These orderings held in all three seeds. Contact type and placement in the network matter for collective connectivity alongside edge count.

2.4 The structural ordering extends across observation shells

Across 0, 2, 4 and 6 nm surface-gap shells, E1H1 retained the largest mean all-agent component and largest-component SV fraction. Its all-agent fraction ranged from 0.4239 to 0.7542, compared

with 0.0131–0.0881 for E0H0. The ordering was consistent across these graph thresholds applied to the same trajectories.

2.5 Stronger assembly accompanies slower short-lag motion

We estimated synapsin mobility from saved positions projected onto the xy plane. Apparent diffusion coefficients came from fits to each agent's time-averaged mean-squared displacement (MSD) over lags of 2.10–3.92 ms. Across-seed means of the agent-mean apparent diffusion coefficient were 0.1403, 0.1339, 0.1152 and 0.0756 $\mu m^2/s$ for E0H0, E0H1, E1H0 and E1H1, respectively (Fig 4). Short-lag mobility was about 46% lower in E1H1 than in E0H0.

The released Hoffmann et al. source data provide an experimental scale reference. They report a mean short-time synapsin coefficient of 0.0189 $\mu m^2/s$ with SVs, versus 0.0253 $\mu m^2/s$ without SVs [8]. The E1H1 model mean is about four times the experimental with-SV mean. Restricting the simulation analysis to the last 60 frames lowers the E1H1 mean to approximately 0.0659 $\mu m^2/s$, still above that reference. The full-trajectory estimates are shown in the main figure; the late-window estimates and their variability are given in S1 Text. Both the experimental and simulated comparisons show lower synapsin mobility in the more assembled setting.

3 Discussion

3.1 Local association and collective organization respond differently

The main result separates vesicle coverage from network connectivity. Electrostatics alone placed synapsin near almost every vesicle, but most vesicles remained in separate groups. Hydrophobic attraction alone gave less synapsin association and brought more vesicles into the largest group. With both interactions active, a much larger network formed.

The pair forces help explain this pattern (table 3). Electrostatics attracts Syn–SV pairs and repels like-species pairs. Hydrophobic attraction acts on Syn–Syn and Syn–SV pairs. E1H0 therefore favored Syn–SV contacts, while E0H1 had more Syn–Syn and SV–SV contacts. With both interactions active, many more synapsin particles were close to several vesicles. Such positions can connect vesicles in the association graph.

Contact number alone did not determine the size of the assembly. E1H0 had more edges and

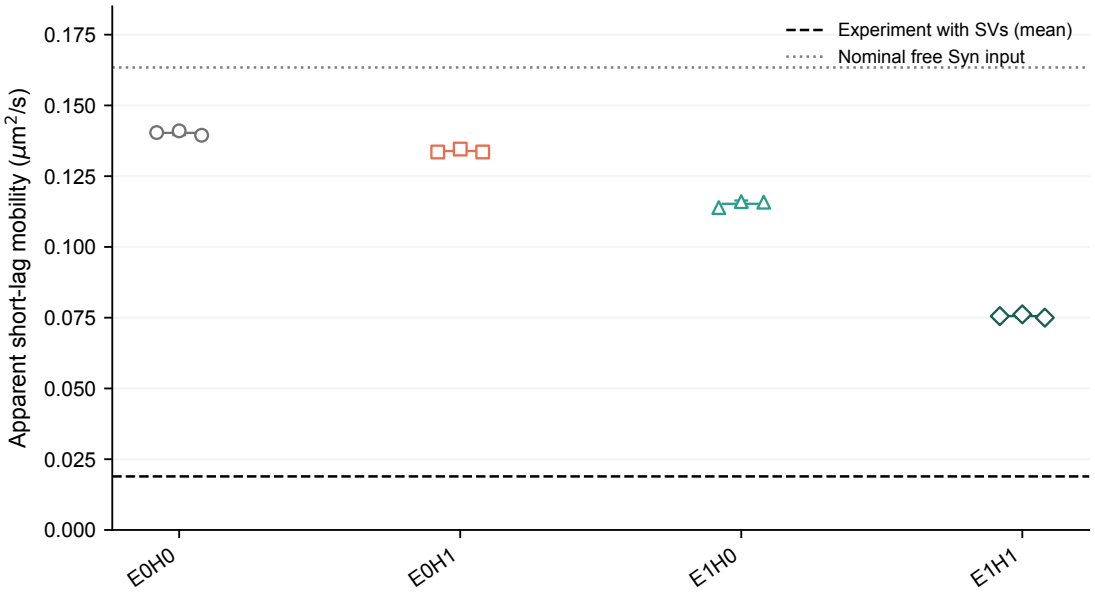


Fig 4: Short-lag synapsin mobility and the experimental reference. Symbols show run means over synapsin agents; marks and bars show the between-seed mean and one SD. The apparent coefficient is one quarter of the free-intercept slope of projected time-averaged MSD over 2.10–3.92 ms. The dashed line is the Hoffmann et al. with-SV mean from released Fig. 2h source data [8]. The dotted line shows the nominal unconfined synapsin diffusion coefficient.

Table 3: Pair-level action of the two interaction terms at positive outer-surface gap.

Pair class	Electrostatic interaction when E1	Hydrophobic interaction when H1
Syn–Syn	Repulsive	Attractive
Syn–SV	Attractive	Attractive
SV–SV	Repulsive	Zero amplitude

Signs describe radial forces for the assigned effective surface potentials. Dispersion attraction acts in all three pair classes and remains enabled in every condition.

more synapsin near multiple SVs than E0H1, yet its largest component contained fewer vesicles. Edges within a group leave the number of connected vesicles unchanged. An edge between groups can join many vesicles at once.

This nonlinear change resembles connectivity transitions studied in random graphs and continuum percolation [17,18]. The additive reference gives a largest-component fraction of 0.1805, while E1H1 reached 0.5528. The relevant experimental comparison would measure vesicle coverage and group connectivity during perturbations of protein–membrane and protein–protein interactions.

3.2 Relation to molecular and particle-level models

The particle model of Alfken et al. links redistribution of vesicle-bound synapsin to adhesion and finite assembly size [11]. Their two-dimensional periodic Langevin model follows proteins on vesicle surfaces. Our model follows freely moving effective synapsin particles inside a three-dimensional compartment and separates electrostatic from hydrophobic interactions. The two models address different routes from local interaction to vesicle assembly.

Residue-level simulations describe charge and hydropathy along protein chains [15]. In repetitive peptides, hydrophobic attraction competes with same-sign electrostatic repulsion [16]. Our two-species model represents this competition for Syn–Syn pairs and includes attraction between oppositely charged Syn–SV pairs. The latter favors local protein association, while larger vesicle assemblies also depend on how the contacts connect separate groups.

Atomistic simulations and experiments connect an ATP-dependent synapsin conformational transition to vesicle binding and oligomerization [14]. At the mesoscale, our contact-network analysis shows how the balance of effective interactions changes the organization of many vesicles.

3.3 Experimental context and biological implications

Reconstitution and neuronal experiments establish synapsin as a regulator of vesicle organization [6, 7]. Scattering measurements show noncompact synapsin–vesicle clusters [12]. Their structure makes connected-component measures relevant alongside cluster size. Studies of synaptophysin tails also support electrostatic interactions in protein–vesicle assembly [19, 20].

Joint activation reduced short-lag synapsin mobility by about 46% relative to E0H0. Hoffmann et al. likewise observed restricted synapsin motion in the presence of vesicles [8]. The simulated

apparent diffusion coefficient was about four times their with-SV mean. A quantitative mobility match would require further calibration.

The distinction between coverage and connectivity may also help interpret synapsin dysfunction. Human *SYN1* loss-of-function variants associated with autism and epilepsy impair vesicle-pool size and trafficking [21]. Oxidation and persulfidation have been linked to different changes in synapsin condensation and neurotransmission [22]. Together, these studies motivate experiments that measure local vesicle association and network connectivity.

3.4 Scope and limitations

The four conditions were compared over 28 ms. E1H1 was still growing at that point, although its growth slowed in the 84-ms continuation (Fig 2C). The contrasts may therefore reflect assembly rates as well as later structure. Three seed blocks gave consistent directions of response, with limited precision for effect sizes. We studied one concentration and compartment volume; other concentrations may change the contrast near a connectivity threshold or saturation.

The agents are spheres with effective surface interactions in a closed compartment. Switching E or H changes several pair classes at once. Pair-specific perturbations would resolve their separate contributions. Graph edges describe proximity. A synapsin near several vesicles is counted as a multivesicle contact; molecular binding is unresolved. The model omits explicit valency, phosphorylation, membrane deformation, active vesicle cycling and hydrodynamic coupling. Numerical checks establish geometric admissibility; convergence of the reported observables at smaller integration steps remains to be assessed. Connectivity and short-lag motion characterize assembly here. A claim of liquid–liquid phase separation would also need evidence for phase coexistence, exchange and system-size behavior.

The mobility reference has different sampling: Hoffmann et al. selected condensate-core tracks, while our estimate includes every synapsin agent and uses a different saved cadence without localization noise. The fourfold difference in apparent mobility is a scale comparison, pending calibration against matched tracking conditions. The interaction switches likewise do not represent specific disease variants or biochemical modifications.

Conclusion

Electrostatic and hydrophobic interactions shape synapsin–vesicle assemblies in different ways. Electrostatics gave broad local coverage, while hydrophobic attraction alone led to greater vesicle participation in the largest group. Together they produced a much larger assembly, more synapsin near multiple vesicles and slower short-lag motion. The results show why vesicle coverage and network connectivity should be measured together: local contacts can be abundant even when vesicle groups remain separate.

4 Methods

4.1 Simulation design and numerical settings

The four force conditions each have three independent seed blocks (9310002–9310004), giving 12 trajectories. Every run contains 200 SVs and 1,807 synapsin agents in a closed $0.1\,\mu\mathrm{m}^3$ compartment. Each trajectory comprises 40,000 integration steps and 200 saved frames. The numerical settings are common to all conditions. S1 Text gives the per-run numerical diagnostics.

4.2 Compartment, size distributions, and population

We represent SVs and synapsin as spherical agents in three dimensions, following the scale of synapsin–vesicle reconstitution and neuronal imaging [6, 8]. The closed spherical compartment has volume $V = 0.10\,\mu\mathrm{m}^3$, corresponding to radius

$$R_{\mathrm{b}} = \left(\frac{3V}{4\pi}\right)^{1/3} = 287.94\,\mathrm{nm}. \tag{2}$$

The scale is informed by three-dimensional hippocampal bouton reconstructions [23]. An SV has membrane-core radius b_i, corona thickness $s_i = 4.8\,\mathrm{nm}$, and outer radius $a_i = b_i + s_i$. Synapsin has $b_i = a_i = 3.972\,\mathrm{nm}$ and $s_i = 0$. The protein corona is interpenetrable [12]; the synapsin radius sets the effective particle size.

SV core radii are independently rejection-sampled from a normal distribution with location $\mu_R = 20.45\,\mathrm{nm}$ and input SD $\sigma_R = 2.5\,\mathrm{nm}$, restricted to $15 \le b_i \le 30\,\mathrm{nm}$. The mean and SD inputs follow the excitatory-SV diameter measurements of Tao et al. [24]; the bounds are informed by cryo-

electron microscopy (cryo-EM) and small-angle X-ray scattering (SAXS) size measurements [25]. These values specify the distribution before truncation. Synapsin counts are obtained from

$$N_{\mathrm{Syn}} = \left\lfloor N_{\mathrm{A}} C_{\mathrm{Syn}} V + \frac{1}{2} \right\rfloor . \tag{3}$$

Here N_{A} is Avogadro's constant, with C_{Syn} in mol/m^3 and V in m^3; $\lfloor x+1/2 \rfloor$ rounds a nonnegative value to the nearest integer. The chosen 30 µmol/L concentration gives $N_{\mathrm{Syn}} = 1807$. There are 200 SVs at the canonical volume. Initial centers are sampled uniformly within active cells and restricted to the particle-accessible sphere. Before the physical trajectory, an overlap-relaxation stage reduces initial core intersections. Its stopping criterion is an overlap below 0.05 nm for three checks, or no overlapping pairs; the stage has a 200-iteration ceiling.

4.3 Linked cells and boundary treatment

The supremum outer SV diameter is $d_{\max} = 2(30 + 4.8) = 69.6\,\mathrm{nm}$. For each bounding-box dimension L_d, the grid uses

$$n_d = \max\{1, \lfloor L_d / d_{\max} \rfloor\}, \qquad h_d = L_d / n_d. \tag{4}$$

The canonical sphere has eight cells per bounding-box axis, with $h_d = 71.9853\,\mathrm{nm}$. Cells are assigned by particle center. The force calculation searches the home cell and two neighboring cell layers. Every candidate pair enters the force calculation; no further radial cutoff is imposed.

The largest center distance at a 6-nm outer-surface gap is $d_{\max} + 6 = 75.6\,\mathrm{nm}$, within the two-layer search distance of 143.9706 nm. A particle center is confined to radius $R_{\mathrm{b}} - a_i$ from the compartment center. Out-of-domain proposals are mapped inward by radial folding; the same boundary map is applied after overlap corrections. The resulting boundary approximates a closed reflecting wall.

4.4 Pair interactions

Let $r_{ij} = \|\boldsymbol{x}_i - \boldsymbol{x}_j\|$, $\boldsymbol{n}_{ij} = (\boldsymbol{x}_i - \boldsymbol{x}_j)/r_{ij}$, and $g_{ij} = r_{ij} - a_i - a_j$. A positive radial force is repulsive. The production force definition is

$$\boldsymbol{F}_i = \sum_{j \in \mathcal{N}_i} \mathbf{1}\{g_{ij} \geq 0\} \left(E f_{ij}^E + H f_{ij}^H + f_{ij}^W \right) \boldsymbol{n}_{ij}. \tag{5}$$

Here $\mathcal{N}_i$ is the linked-cell candidate set and $E, H \in \{0, 1\}$ switch the electrostatic and hydrophobic terms. Van der Waals attraction remains active in every condition. At negative outer gap, the surface-force terms vanish and geometric correction controls core overlap.

The electrostatic kernel is a reduced-potential, screened sphere interaction:

$$f_{ij}^E = k_\mathrm{B} T \frac{a_i a_j}{\ell_\mathrm{B}} \varphi_i \varphi_j e^{-\kappa g_{ij}} \left(\frac{1}{r_{ij}^2} + \frac{\kappa}{r_{ij}} \right), \tag{6}$$

$$\varphi_i = \frac{e\psi_i}{k_\mathrm{B} T}, \quad \ell_\mathrm{B} = \frac{e^2}{4\pi\epsilon_0 \epsilon_r k_\mathrm{B} T}, \quad \kappa^2 = \frac{2e^2 N_\mathrm{A} (1000 I)}{\epsilon_0 \epsilon_r k_\mathrm{B} T}. \tag{7}$$

The last expression uses SI quantities and I in mol/L; lengths are then converted to nanometers. At $T = 310.15\,\mathrm{K}$, $\epsilon_r = 74.2$, and $I = 0.15\,\mathrm{mol/L}$, $\kappa^{-1} = 0.77886\,\mathrm{nm}$. SV and synapsin effective potentials are respectively -22.5 and $+16\,\mathrm{mV}$. Negative isolated-SV zeta potentials motivate the former [26]; the latter represents an effective positive protein charge scale. The kernel approximates screened electrostatics in a linearized-Poisson–Boltzmann/Yukawa form.

Hydrophobic attraction is

$$f_{ij}^H = -\frac{\epsilon_{ij} k_\mathrm{B} T}{\lambda_H} e^{-g_{ij}/\lambda_H}, \qquad \lambda_H = 0.5\,\mathrm{nm}. \tag{8}$$

This follows from $U_{ij}^H = -\epsilon_{ij} k_\mathrm{B} T e^{-g_{ij}/\lambda_H}$ outside contact. An exponential kernel has a surface-force precedent [27]; its length and pair amplitudes specify the effective short-range attraction. The SV–SV hydrophobic amplitude is zero.

For the fixed dispersion background, the membrane/core radii are used:

$$f_{ij}^W = -\frac{A\, b_i b_j}{6(b_i + b_j)\widehat{h}_{ij}^2}, \qquad \widehat{h}_{ij} = \max\{r_{ij} - b_i - b_j, h_0\}, \tag{9}$$

where $A = 5 \times 10^{-21}\,\mathrm{J}$ and $h_0 = 0.3\,\mathrm{nm}$. This non-retarded Hamaker–Derjaguin approximation is informed by dispersion theory [28]; it uses a common effective aqueous background constant. The dispersion surfaces are defined by core radii, with lower-gap regularization to bound the attraction near contact.

4.5 Mobility and discrete stochastic dynamics

Independent-particle Stokes drag gives

$$\gamma_i = 6\pi\eta_{\mathrm{eff}} a_i, \qquad M_i = \gamma_i^{-1}, \qquad D_i = k_{\mathrm{B}} T M_i. \tag{10}$$

Brownian particle updates use diagonal mobility [29]. The effective friction input is $\eta_{\mathrm{eff}} = 0.35\,\mathrm{Pa\,s}$, selected with reference to published mobility scales [8, 30]. The nominal unconfined diffusion coefficients are $0.1634\,\mu\mathrm{m}^2/\mathrm{s}$ for synapsin and $0.02571\,\mu\mathrm{m}^2/\mathrm{s}$ for an SV with $a_i = 25.25\,\mathrm{nm}$.

At each Euler–Maruyama step,

$$\boldsymbol{d}_i^n = M_i \boldsymbol{F}_i(\boldsymbol{x}^n)\Delta t, \qquad c_i^n = \min\left\{1, \frac{a_i/4}{\|\boldsymbol{d}_i^n\|}\right\}, \tag{11}$$

$$\boldsymbol{y}_i = \mathcal{B}_i\left(\boldsymbol{x}_i^n + c_i^n \boldsymbol{d}_i^n + \sqrt{2D_i\Delta t}\,\boldsymbol{\xi}_i^n\right), \qquad \boldsymbol{\xi}_i^n \sim N(\boldsymbol{0}, \boldsymbol{I}_3). \tag{12}$$

For zero drift, $c_i^n = 1$. The limiter acts on deterministic displacement, while the Brownian increment remains Gaussian. $\mathcal{B}_i$ is the boundary map. The next state is the result of mobility-weighted iterative overlap correction applied to $\boldsymbol{y}$. Pair admissibility is

$$r_{ij} \geq b_i + b_j - \delta_0, \qquad \delta_0 = 0.5\,\mathrm{nm}. \tag{13}$$

Thus permitted outer overlaps are 0.5 nm for Syn–Syn, 5.3 nm for Syn–SV, and 10.1 nm for SV–SV. Corrections are accumulated simultaneously, averaged over each particle's violating contacts, reflected at the wall, and repeated with updated cells. The residual tolerance is 0.005 nm, with early stopping and a 128-iteration safety ceiling.

The fixed integration step is $\Delta t = 0.0007\,\mathrm{ms}$. Each trajectory has 40,000 physical moves over 28 ms; snapshots are scheduled every 200 steps, or 0.14 ms. Forces are evaluated at the stored state

before movement, aligning saved forces and positions.

4.6 Structural graphs and observables

A nonperiodic undirected graph is constructed independently at each stored frame using $A_{ij}^{(\ell)} = \mathbf{1}\{g_{ij} \le \ell\}$, $i \neq j$. The primary outer-surface shell is $\ell = 2\,\mathrm{nm}$, with $0, 2, 4, 6\,\mathrm{nm}$ sensitivity shells. Edges represent geometric association. Let $\mathcal{C}_f$ be the connected components at frame f, $n_c = |c|$, and c_f^* the component with the largest total agent count. The species-resolved structural observables are

$$L_{\mathrm{SV},f} = \frac{|c_f^* \cap \mathrm{SV}|}{N_{\mathrm{SV}}}, \tag{14}$$

$$Q_{\mathrm{Syn},f} = \frac{\sum_{c \in \mathcal{C}_f : n_c \ge 2} |c \cap \mathrm{Syn}|}{N_{\mathrm{Syn}}}. \tag{15}$$

Thus $L_{\mathrm{SV},f}$ counts vesicles in the component selected by total agent count, and $Q_{\mathrm{Syn},f}$ counts synapsin in any component containing at least two agents, including Syn–Syn groups. The primary structural observable is the all-agent largest component fraction

$$L_{\mathrm{all},f} = \frac{|c_f^*|}{N_{\mathrm{SV}} + N_{\mathrm{Syn}}}, \tag{16}$$

while $Q_{\mathrm{Syn},f}$ and $L_{\mathrm{SV},f}$ describe local protein participation and collective vesicle participation, respectively.

Each outcome is averaged over the last $m = \max\{1, \lceil 0.30F \rceil\}$ of the F stored frames:

$$Y_{o,s} = \frac{1}{m} \sum_{f=F-m+1}^{F} O_{o,s,f}. \tag{17}$$

Here o indexes the observable, s the seed and f the saved frame; the force-condition index is suppressed. Each run is summarized before values are compared across seeds. For these trajectories, the late window covers frames 141–200 (19.74–28 ms).

4.7 Force-factorial design

The factorial design fixes ionic strength at $I = 0.15\,\mathrm{mol/L}$ and nominal synapsin concentration at $30\,\mu\mathrm{mol/L}$. It includes all four combinations of the binary interaction switches, $E, H \in \{0, 1\}$. H1 uses the reference amplitudes $\epsilon_{\mathrm{SynSyn}} = 1.50$ and $\epsilon_{\mathrm{SynSV}} = 1.00$, corresponding to well depths of $1.50 k_{\mathrm{B}}T$ and $1.00 k_{\mathrm{B}}T$, respectively, with $\epsilon_{\mathrm{SVSV}} = 0$; H0 sets all hydrophobic amplitudes to zero. E1 uses the electrostatic kernel of eq. (6) and E0 sets that kernel to zero. In every condition, the same van der Waals law and its parameters remain active.

Three seed blocks, 9310002–9310004, were completed in all four conditions. An earlier four-run pilot with seed 9310001 informed feasibility and the selection of structural observables. The 12 later trajectories provide the comparisons reported here. The 2-nm primary shell and the three sensitivity shells were selected before those runs. Fixed inputs are listed in table 4.

Table 4: Fixed inputs of the four-condition force comparison.

Quantity	Value	Model role
Integration step	0.0007 ms	Fixed in every condition
Duration	28 ms	40,000 physical steps
Saved-frame spacing	0.14 ms	200 saved frames per completed run
Temperature	310.15 K	Shared setting
Effective viscosity	0.35 Pa s	Literature-informed effective friction
Relative permittivity	74.2	Solvent approximation
Volume	$0.10\,\mu\mathrm{m}^3$	Closed spherical compartment
Synapsin concentration / count	$30\,\mu$M / 1,807	Same in all conditions
Vesicle count	200	Same in all conditions
Vesicle core-radius input	20.45 nm	Bounds: 15–30 nm
Core-radius SD input	2.5 nm	Truncated heterogeneous input
Corona thickness	4.8 nm	Effective interpenetrable shell
Synapsin effective radius	3.972 nm	Coarse-grained input
SV / Syn surface potentials	-22.5 / $+16$ mV	Effective surface-potential inputs
Hamaker constant	5×10^{-21} J	Fixed van der Waals background
Surface regularization	0.3 nm	Numerical/force definition
Hydrophobic decay length	0.5 nm	Effective short-range input
H1 well depths Syn–Syn / Syn–SV	1.50 / 1.00 $k_{\mathrm{B}}T$	Syn–SV and Syn–Syn only; H0 sets zero
Core overlap / residual	0.5 / 0.005 nm	Distinct geometric quantities
Projection iteration cap	128	Early stopping enabled
Neighbor depth	Two cell layers	Candidate-pair force evaluation
Primary analysis shell	2 nm	Sensitivity: 0, 2, 4, 6 nm

E and H vary through the binary switches; the remaining inputs are held fixed.

4.8 Post hoc mobility analysis and experimental reference

The released source-data workbook for Hoffmann et al. reports synapsin motion in the presence of isolated vesicles and in living neurons [8]. Its Fig. 2h summarizes individual apparent diffusion coefficients evaluated over 2–4 ms using 1000-Hz tracking. We use the released with-SV mean as the external scale reference in the main figure.

For each simulated synapsin agent i, coordinates are converted from nm to μm and projected as $\boldsymbol{z}_{i,f} = (x_{i,f}, y_{i,f})$. With $F = 200$ and $\delta t_{\mathrm{save}} = 0.14$ ms, the time-averaged squared displacement at lag k is

$$\widehat{\mathrm{MSD}}_i(k) = \frac{1}{F-k} \sum_{f=1}^{F-k} \|\boldsymbol{z}_{i,f+k} - \boldsymbol{z}_{i,f}\|^2. \tag{18}$$

We fit $\alpha_i + \beta_i \tau$ by ordinary least squares with a free intercept, using all 14 native lags between 2.10 and 3.92 ms and time in seconds. Then $\widehat{D}_i = \beta_i/4$ has units of $\mu\mathrm{m}^2/\mathrm{s}$. The run summary is the mean of $\widehat{D}_i$ over 1,807 synapsin agents. We then report the mean and SD across the three runs in each condition. The estimator is also applied to the final 60 frames to assess window sensitivity. S1 Text gives the agent-level distribution and fitted-slope diagnostics.

4.9 Post hoc species-resolved contact analysis

For every saved frame, the same 2-nm association graph is reconstructed from recorded centers and outer radii. Each unordered edge is counted once and classified as Syn–Syn, Syn–SV or SV–SV. Let $\mathcal{S}$ and $\mathcal{V}$ be the Syn and SV index sets. For Syn agent $i \in \mathcal{S}$, let

$$d_i^{\mathrm{SV}}(t) = \sum_{j \in \mathcal{V}} \mathbf{1}\{\|\boldsymbol{x}_i(t) - \boldsymbol{x}_j(t)\| \le a_i + a_j + 2\,\mathrm{nm}\}.$$

The fractions with at least one or at least two SV neighbors are $N_{\mathrm{Syn}}^{-1} \sum_{i \in \mathcal{S}} \mathbf{1}\{d_i^{\mathrm{SV}} \ge 1\}$ and $N_{\mathrm{Syn}}^{-1} \sum_{i \in \mathcal{S}} \mathbf{1}\{d_i^{\mathrm{SV}} \ge 2\}$. The reciprocal SV coverage counts SVs with at least one Syn neighbor, divided by N_{SV}. Counts and fractions are first averaged within each run over its last 60 frames, and only then summarized across seeds. The analysis uses the same saved trajectories and 2-nm shell as the structural comparison.

4.10 Seed-level statistical analysis

Each complete trajectory is one replicate. Observations of frames, agents and clusters are nested within it [31]. The three seed identifiers define blocks across conditions; their role is to pair run-level comparisons.

Each structural observable is first averaged over the final 60 frames of a trajectory. We report the mean and sample SD of the three run-level values for each condition. For a comparison of conditions A and B, the paired difference is $d_s = Y_{B,s} - Y_{A,s}$; its mean and sample SD summarize the three seed-wise differences. The E-by-H interaction is calculated within each seed block using eq. (1). Individual seed values are shown in the figures and S1 Text. We use the size and consistency of paired differences to interpret the results.

4.11 Use of AI

The authors used ChatGPT (OpenAI) for language revision and assistance with code debugging, and Claude (Anthropic) for feedback on early drafts. The authors checked the code and manuscript and take responsibility for the final content.

Author contributions

Shokoofeh Akbari: Conceptualization, Methodology (mathematical formulation), Software, Investigation, Formal analysis, Data curation, Visualization, and Writing–original draft.
Oleg Shupliakov: Conceptualization (biological and LLPS input) and Writing–review and editing.
Yousef Jamali: Conceptualization, Methodology (scientific interpretation), Supervision, Funding acquisition, and Writing–review and editing.

Data and code availability

The numerical data underlying the figures and tables are provided in the Supporting Information. The reproducibility package contains the simulator and the 12 condition configurations, together with analysis scripts, processed per-seed results, numerical diagnostics and trajectory checksums. It is maintained at `https://github.com/BioMathLab-TMU/simulationsv-electrostatic-hyd`

rophobic-response, release v1.0.0. The package is publicly available and includes the input data and scripts required to reproduce the reported results. Original author-generated code is licensed under MIT; author-generated data and tables are licensed under CC BY 4.0. Third-party code and experimental data retain their original licenses. The independent experimental source data of Hoffmann et al. are available at https://doi.org/10.6084/m9.figshare.24179445.v1.

Competing interests

The authors declare no competing interests.

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Supporting information captions

S1 Text. Supporting methods and numerical results. Per-seed structural endpoints, paired primary-interaction estimates, association-shell sensitivity, interface definitions and complete run-level interface outcomes, together with statistical and software validation details. (PDF)

S1 Data. Numerical data underlying the figures and tables. Per-seed structural observables, paired contrasts, graph-shell sensitivity, time-series inputs, interface diagnostics and mobility comparison values. (ZIP)